Ibn Tofail University

Rigid body mechanics

Problem Set II

Exercise 1:

Determine the number of degrees of freedom of the following systems:

1. Two material points M_1 and M_2 moving freely on a plane while maintaining a constant distance between them.
2. A sphere free to move in space.
3. A sphere free to move on a plane.
4. A sphere free to move on a line.
5. A rod free to move on a plane.
6. A rod with one fixed end moving on a plane.
7. A system of two articulated rods with one fixed end.
8. A hoop free to move on a plane.
9. A hoop free to move on a line.

Correction

Exercise 2:

Let S be a perfect rigid body in motion with respect to a fixed reference frame T. $\vec{\Omega}$ is the instantaneous rotation vector of S with respect to T.

Show that at a given instant, the projection of the velocities of the points of the solid on the axis $\vec{\Omega}$ is constant.

Correction

Exercise 3:

A rectangular plate $ABCD$ moves with respect to a direct orthonormal frame $R(O, X, Y, Z)$ such that A remains coincident with point O and side AB remains in the plane XOY.

1. Determine the number of degrees of freedom of the plate.
2. What is the rotation vector $\vec{\Omega}$ of the plate with respect to R?

Correction

Exercise 4:

A hoop S of center C and radius r moves in the plane OX_1Y_1 of a reference frame $R_1(O, X_1, Y_1, Z_1)$ while remaining in contact with the axis OX_1 at point I, $\overrightarrow{OI} = x\vec{i}_1$. The frame $R_1(O, X_1, Y_1, Z_1)$ is in motion with respect to the fixed reference frame $T_1(O_1, X_1, Y_1, Z_1)$ such that $\overrightarrow{O_1O} = y\vec{j}_1$.

Let M be a point of the solid S. We set $(\vec{i}_1, \overrightarrow{CM}) = \theta$. We attach to the solid a frame $T(C, X, Y, Z)$ with basis $(\vec{i}, \vec{j}, \vec{k})$ such that the X axis passes through point M.

1. Determine the number of degrees of freedom of the solid.
2. Express the instantaneous rotation vector of the hoop with respect to T_1.
3. Can there be pivoting of the hoop with respect to R?
4. Determine the sliding velocity of the hoop with respect to the axis OX_1.
5. Calculate $\vec{V}(I/R)$ and $\vec{V}(I/S)$ then find again the sliding velocity of the hoop with respect to the axis OX_1.
6. Calculate the acceleration $\vec{\gamma}(M/T_1)$. Deduce the accelerations $\vec{\gamma}_S(I/T_1)$ and $\vec{\gamma}_S(J/T_1)$ where J is a point diametrically opposite to I on S.

Correction

Exercise 5:

Consider a homogeneous sphere $S(G, a)$, of center G, mass m, and radius a, moving on the fixed horizontal plane (O, X_0, Y_0) of a direct orthonormal fixed frame $R_0(O_0, X_0, Y_0, Z_0)$. Let $R_S(G, X, Y, Z)$ be the direct orthonormal frame attached to the sphere. I is the contact point between the sphere S and the plane (O, X_0, Y_0). We denote by ψ, θ, and φ the three Euler angles of S/R_0 and x, y, and z the coordinates of G. We assume that S rolls without slipping on the axis (O, X_0).

1. Euler angles

a) Identify the relative Euler frames R_1 and R_2. Deduce the angular variables.

b) Parameterize the sphere S.

c) Determine the expression of the instantaneous rotation vector of S with respect to R_0.

2. Kinematics of the contact point

a) Express the non-pivoting condition of S with respect to R_0.

b) Express the non-rolling condition of S with respect to (O, X_0). Show that one of the obtained equations is semi-holonomic.

c) Calculate the velocities of the different contact points: Point of the sphere $I \in S$; point of the material plane $I \in (O, X_0, Y_0)$; and geometric point $I \in E$.

d) Give the no-slip condition of S with respect to (O, X_0, Y_0). What is the number of degrees of freedom of S in its motion with respect to R_0?

Correction

Exercise 6:

A horizontal disk S_2, of center O_1 and radius R, is rotated about the axis O_1Z_1 of the reference frame $T_1(O_1, X_1, Y_1, Z_1)$ assumed fixed, with angular velocity ω, by a disk S_1, of center C and radius r. S_1 rolls without slipping on the contour of disk S_2, with angular velocity Ω, about the rod AC which is parallel to the plane of S_2. This rod rotates about the axis AZ_1 with angular velocity ω_0 with respect to S_2.

1. Write the expressions for the rotation velocity vectors $\vec{\Omega}(S_1/AC)$, $\vec{\Omega}(AC/S_2)$, and $\vec{\Omega}(S_2/T_1)$.

2. By exploiting the rolling without slipping condition of S_1 on S_2, deduce a relation between the angular velocities Ω, ω_0, and ω.

3. Express the velocity of point J of S_1, which is diametrically opposite to I. What does this velocity become if disk S_2 is stationary?

Correction